

Saam Haghighat-Grami

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Roswell, GA

Interests – Autonomous, Mobile, Collaborative Robotics

Graduating December 2026

LinkedIn – saam-haghighat-grami

GitHub – Saam-Grami

Portfolio – saam-grami.github.io

Active Security Clearance

EDUCATION

- Kennesaw State University** *January 2022 – May 2022, January 2024 – Present*
Mechatronics Engineering, B.S. – Graduation December 2026 CGPA: 4.00
- Marine Corps Communication Electronics School** *August 2022 – October 2023*
Ground Radio Repair School CGPA: 4.00
- Chattahoochee Technical College** *August 2020 – December 2021*
General Education (Undecided Major) CGPA: 3.80

RESEARCH & ENGINEERING EXPERIENCE

- River Debris Trajectory Prediction using Machine Learning** *June 2026 - August 2026*
Undergraduate Researcher – advised by Dr. Matt Marshall
 - Overview:** Independently designing a real-time, edge-deployed system that detects, tracks, and forecasts the trajectories of floating river debris on an NVIDIA Jetson Orin Nano.
 - Approach:** Replacing linear Kalman-filter motion models, which degrade in turbulent flow, with a class-conditioned CNN and LSTM-DNN whose Gaussian uncertainty head gives calibrated multi-step forecasts over YOLO26 detections.
 - Publication:** First-author manuscript in preparation, targeting IEEE SoutheastCon 2027.
- MEAP – Mass Estimation Autonomous Pathfinding for Search and Rescue** *November 2025 - Present*
Robotics Software Architect & Project Coordinator – AgriSys Lab, Dr. Muhammad Hassan Tanveer
 - Overview:** Designing a UAV–UGV collaborative search-and-rescue system: the UAV surveys the environment while the UGV chooses advantageous traversable obstacles to clear with an onboard manipulator to shorten the path to each objective.
 - Contribution:** Leading the UAV autonomy stack: flight motion framework, aerial map stitching, and the pipeline that turns survey passes into one searchable map. Also leading the ML pipeline estimating obstacle mass from sensor-fused SLAM point clouds to determine what the manipulator can safely move.
 - Publication:** First-author manuscript in preparation, targeted for submission December 2026.
- DARB – Drone Arm Robotic Backpack (Personal Project)** *May 2025 - Present*
Sole Designer and Builder
 - Overview:** Designing and building an autonomous drone docking system: a robotic arm on a wearable backpack platform captures a drone mid-flight, stows it, and relaunches it.
 - Approach:** Built a custom airframe that uses computer vision to align to IR LED markers on the platform, triggering arm capture once orientation converges.
 - Hardware:** Custom airframe on a Matek H743 flight controller running ArduPilot, with encoder-driven NEMA 23 steppers actuating the capture arm.
- SoilBus – Soil Monitoring Probe Network** *August 2025 - December 2025*
Lead Software Developer – AgriSys Lab, Dr. Muhammad Hassan Tanveer
 - Overview:** Built a hub-less soil-monitoring mesh network for farmers: probes auto-discover neighbors and any single probe can offload the entire network's data.
 - Results:** Used breadth-first search for inter-probe routing; validated across 5 physical nodes, holding to 26 simulated nodes before degradation.
 - Publication:** SoilBus: An Ad Hoc Soil Sensor Network for Precision Agriculture.

INDUSTRY & MILITARY EXPERIENCE

- EOSYS Industrial and Manufacturing System Integrator** *May 2025 – August 2025*
COOP Marietta GA
 - HMI tooling:** Built a Human Machine Interface display generator for P&IDs.
 - Controls:** Designed and modified PLC programs.
- United States Marine Corps Reserve** *August 2022 - Present*
Ground Radio Repair Specialist Marietta GA
 - Radio systems:** Troubleshoot, maintain, and operate USMC radio systems in time-sensitive conditions; active security clearance.
 - Squad Leader:** 4th Combat Readiness Regiment; training, accountability, and readiness of a squad of Marines.

TECHNICAL SKILLS

Perception & ML: PyTorch, TensorFlow, YOLO, CNNs, LSTMs, OpenCV, ORB-SLAM3, sensor fusion, TensorRT
Autonomy & Robotics: ROS 2, Nav2, ArduPilot, PX4, Mission Planner, path planning, UAV/UGV integration
Embedded & Hardware: ESP32, Arduino, Raspberry Pi 5, Jetson Orin Nano, OpenMV, LiDAR, RGB-D cameras, IMUs, ad hoc networking
Manipulators & Automation: Denavit-Hartenberg parameters, forward and inverse kinematics, trajectory planning, Ladder Logic, TIA Portal, PLCs, HMI development
Languages & Tools: C++, Python, MATLAB, LaTeX, Linux/WSL2, Git, VS Code, SolidWorks, Onshape